

POAMS: LECTURE 7
UNIFORM MOTION: LENGTH CONTRACTION, MINKOWSKI
SPACE-TIME AND LORENTZ TRANSFORMATIONS

Having derived the time-dilation formula for uniform motion in our Normal Realist alternative to Einstein's Special Relativity, in this lecture I will examine some further consequences of this, including length contraction, Minkowski space-time and Lorentz transformations, which are all standard results in Special Relativity. The lecture is based on some of 'Light Speed...' Chapters 3 and 4.

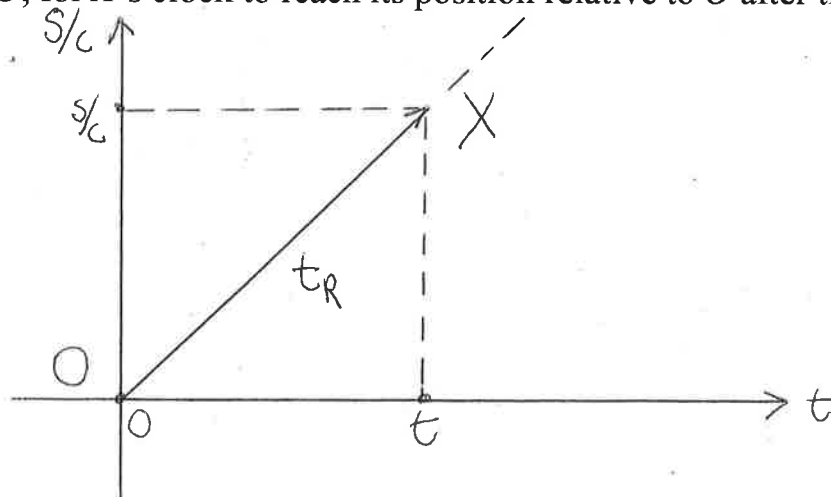
1. Time-dilation in uniform motion

Once again, as a reminder, our Normal Realist postulates for the study of *uniform motion* are:

Postulate 1: *The laws of Physics are the same in all inertial frames.*

Postulate 2: *Observational distance and time have a constant ratio of units, c , for all inertial observers sharing that same conventional choice of units.*

Consider the case of a body X in uniform motion relative to an observer O , as studied in the last two lectures. Body X travels a distance s measured in metres, in a time t , recorded in seconds by X 's clock, relative to O . In keeping with Postulate 2, this distance travelled by X relative to O can be recorded in seconds by the use of the conversion factor c . We then have the *two-dimensional time* diagram, as shown below. In the figure, the two axes (labelled s/c and t) are both measured in units of seconds. Bearing in mind that the line segment OX represents the path of X 's clock relative to O , it follows that the length of OX in this second graph is the time, t_R , as recorded by O , for X 's clock to reach its position relative to O after time t .



It follows by Pythagoras's Theorem that

$$t_R^2 = t^2 + (s/c)^2 \quad (1)$$

Now the speed, v , of X relative to O (that is, relative to O 's clock) is given by $v = s/t_R$, so that Equation (1) becomes

$$t_R^2 = t^2 + (vt_R/c)^2$$

Then after re-arranging this becomes,

$$t = (1 - v^2/c^2)^{1/2} t_R \quad (2)$$

This equation implies that O 's time is dilated relative to X 's time and that a moving clock runs *slower* relative to a stationary observer's clock. Equation (2) is the standard formula for *time-dilation* which appears in Special Relativity. Here t_R is the *observational time* as recorded by the observer of the motion and t is the *proper time* that the observer sees in his telescope/spectroscope registered by the moving body itself.]

[End of recap]

2. Length contraction

In the scenario just described, the distance s travelled by X , as observed by O in the time t_R , is given by $s = vt_R$. Meanwhile, for that same distance X 's clock, as O observes, registers time t , so that the distance travelled by X , which for O is $s = vt_R$ is, for X , only $s_X = vt$. It then follows by Equation (2) that

$$\begin{aligned} s_X/v &= (1 - v^2/c^2)^{1/2} s/v \\ \therefore s_X &= (1 - v^2/c^2)^{1/2} s \end{aligned} \quad (3)$$

From Equation (3) it follows that $s_X < s$ for $v < c$, so that distance travelled by a moving body, relative to that body, is shortened relative to the distance it travels, relative to a stationary observer. This result is related to the *length-contraction effect*, as predicted by Special Relativity. It is essentially the *Lorentz-Fitzgerald contraction effect* dreamed-up by the

classical physicists to explain the null results of the Michelson-Morley experiments, with no logical foundation. [See *Lecture 4*]. We have shown here that this phenomenon follows logically from the time-dilation effect.

3. Minkowski space-time

From now on, we shall adopt the standard notation and denote the *observational time* as recorded by an observer's clock by t , rather than by t_R , as in previous sections. We shall denote the *proper time*, recorded by a body, X , moving relative to an observer O , by τ , instead of t . [*Hopefully, this will not lead to any confusion.*] Remember this proper time is *independent* of any observer recording the motion of X .

In view of the time-dilation formula, which, in our newly adopted notation, reads

$$\tau = (1 - v^2/c^2)^{1/2} t \quad (2a)$$

the clocks of different observers run at relatively different rates. Hence, it is clearly not adequate to talk of a particular observation of O 's occurring at a particular point in space relative to O ; rather that particular observation must be specified not only by its position, but also by the time t it happens relative to O . An observation occurring at a point in space with Cartesian coordinates (x, y, z) and at a time t relative to an observer O , is called the **event** with coordinates (t, x, y, z) . [*Note that it is not universal convention to have the t coordinate first here; some authors put it last.*] In this context, the observational time t is often called O 's **coordinate time**. The set of all events (over all possible values of t, x, y and z) is known as **space-time**, the hyphen conveying the fact that space and time are intimately connected, in the way we have already described.

Consider now any clock X moving with uniform velocity (*i.e.* with constant speed in a fixed direction) relative to an observer O , which passes O at observational time $t = 0$. Then relative to O 's spatial Cartesian coordinates x, y , and z , after a passage of time t relative to O , X travels a radial distance, r , where, by Pythagoras's Theorem,

$$r^2 = x^2 + y^2 + z^2 \quad (4)$$

It follows by Equation (1), bearing in mind our new notation, that the proper time τ recorded by clock X at this time is given by

$$t^2 = \tau^2 + (r/c)^2 \quad (1a)$$

that is,

$$-c^2\tau^2 = -c^2t^2 + r^2$$

and so by Equation (4),

$$-c^2\tau^2 = -c^2t^2 + x^2 + y^2 + z^2 \quad (5)$$

Since both τ and c in Equation (5) are invariant with respect to all observers, it follows that the right-hand-side of this equation provides an invariant measure for all inertial observers. This is known, in this context, as the **separation** or **interval** between two events, as in Special Relativity. The set of all events, together with this invariant quantity, provides a mathematical model for studying uniform motion, known as **Minkowski space-time**. [*Minkowski was one of Einstein's school teachers, who referred to him as a 'lazy dog'!*]

More generally, suppose that the clock X moves with *non-constant velocity*, $\mathbf{v}(t)$, relative to observer O and passes O at observational time $t = 0$. In this case, it follows by Equation (2a) that the relatively moving clock X records proper time τ given by

$$\tau = \int_0^t (1 - v^2(t)/c^2)^{1/2} dt \quad (6)$$

where $v(t) = \|\mathbf{v}(t)\|$ is the speed of X relative to O . The **velocity** of X , relative to O 's spatial Cartesian coordinates x , y and z , is defined by

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$

[*velocity is the rate of change of position*] Hence, Equation (6) gives

$$\frac{d\tau}{dt} = (1 - v^2(t)/c^2)^{1/2} \Rightarrow \left(\frac{d\tau}{dt}\right)^2 = 1 - \|\mathbf{v}(t)\|^2/c^2$$

$$\Rightarrow c^2 \left(\frac{d\tau}{dt}\right)^2 = c^2 - \left(\frac{dx}{dt}\right)^2 - \left(\frac{dy}{dt}\right)^2 - \left(\frac{dz}{dt}\right)^2$$

so that,

$$-c^2 d\tau^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2 \quad (7)$$

Once again, since both τ and c in Equation (7) are invariant with respect to all observers, it follows that the right-hand-side of this equation provides an invariant measure for all inertial observers. This quantity is the infinitesimal version of the right-hand-side of Equation (5) and is known as the **elementary separation (interval)** between two events. Expression (7) is also known as the **metric** for Minkowski space-time, and forms the basis of the study of relativistic kinematics and dynamics.

It is to be noted here that our Normal Realistic approach to relative uniform motion has a big advantage over standard Special Relativity in the sense that we are able to construct Minkowski space-time directly from the time-dilation formula. Moreover, in this approach, Minkowski space-time is not simply a convenient mathematical model for studying relative uniform motion but, rather, a direct expression of the fundamental physical phenomenon of time-dilation. By contrast, in standard Special Relativity, Minkowski space-time is presented as a consequence of the fact that the (elementary) separation of events is invariant with respect to all special Lorentz transformations. Moreover, the invariance of the separation of events is not, generally, presented as a consequence of the invariance of proper time; rather, the invariance of proper time is presented as a consequence of the invariance of the separation of events.

4. The special Lorentz transformation

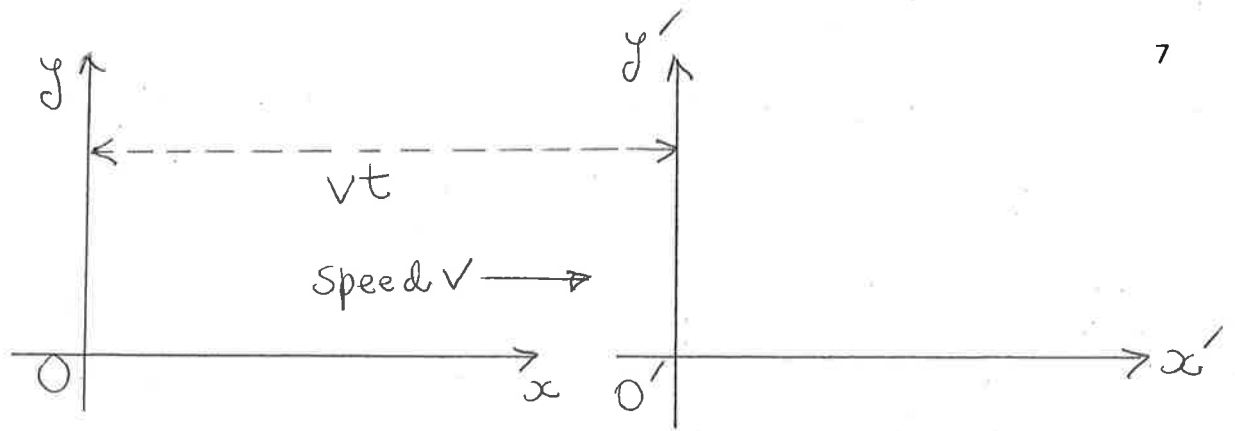
Recall that Normal Realism takes full account of the empirical fact that there is no direct or instrumental way in which the world can be described from two or more separate places at once. The nearest we get, in Physics,

to that objective, decentralised view of the world is *via* the *special Lorentz transformation*. This is why, in standard Special Relativity, much use is made of that transformation, which enables one, in theory, to project one's own observational perspective to those of other observers in the way the English philosopher Alfred J. Ayer (1910-1989) described:

'Places are places in a visual field and the times those that furnish the temporal order of experience... In this way, the framework is set for the description of visual appearances... We can identify a sense field by reference not only to its own character but also that of its neighbours, and if necessary to that of their neighbours and so on.'

In this section, I will demonstrate how the special Lorentz transformation, which relates spatial positions and times, as recorded by different *inertial* observers, may be derived in Normal Realist terms. I shall derive this transformation directly from the time-dilation formula, Equation (2a), without any of the 'electrodynamical' rationale of the standard derivation in Special Relativity, and in particular, without Einstein's second postulate that light travels in *vacuo* at a speed which is constant relative to all differently moving observers.

Consider two observers, O and O' [*instead of X for ease of notation*], each situated at the origin of a set of Cartesian axes labelled x, y and z , and x', y' and z' respectively. Suppose that O' moves along the x -axis in the positive direction with *constant* speed v relative to O , such that the x', y' and z' axes remain parallel to the x, y and z axes respectively. Let t be some passage of time as recorded by a clock in O 's observational frame and let t' be the corresponding passage of time as recorded by a clock in O' 's observational frame. Suppose also that the observers O and O' synchronise their clocks when they coincide at $t = 0$, so that when $t = 0$ at $(0, 0, 0)$ relative to O , we have $t' = 0$ at $(0, 0, 0)$ relative to O' . This situation is indicated in the figure below, in which the z dimension is suppressed. We shall call this the *standard situation* between the inertial observers O and O' .



This diagram shows the position of O' 's frame relative to O 's, after time t , as recorded by O . According to classical physics, after this time relative to O has elapsed,

$$x' = x - vt, \quad y' = y, \quad z' = z, \quad t' = t \quad (8)$$

Equations (8) together are known as the **Galilean transformation**. However, since time is not an absolute quantity, as has been demonstrated, this transformation cannot give the correct relationship between an event relative to two different inertial frames.

In general, by hypothesis, observer O' moves with constant velocity relative to O , so that objects which move with constant velocity relative to O must move with constant velocity relative to O' . It then follows from our initial condition concerning the times t and t' that the most general transformation which maps (x, t) to (x', t') under these conditions must be *linear*. By hypothesis, after time t relative to O , O' has moved a distance $x = vt$, as shown in the figure above, so that, for some constant γ to be determined,

$$x' = \gamma(x - vt) \quad (9)$$

It follows from Equation (9) and our Postulate 1 that since O travels at speed v in the negative x' direction relative to O' , then

$$x = \gamma(x' + vt') \quad (10)$$

Substituting Equation (9) into Equation (10) gives

$$x = \gamma(\gamma x - \gamma vt + vt')$$

Dividing by γv and then re-arranging gives

$$t' = \frac{x}{\gamma v} - \frac{\gamma x}{v} + \gamma t$$

$$\Rightarrow t' = \gamma(t - (1 - 1/\gamma^2)x/v) \quad (11)$$

Equations (9) - (11) are the same equations which appear in Special Relativity. However, whereas Special Relativity resorts to the concept of ‘travelling light signals’ in order to deduce the value of γ in Equation (11), in our approach this value is obtained directly by using the time-dilation formula, Equation (2a). Note that Equations (9) and (11) include the Galilean transformation, Equations (8), as a special case, when $\gamma = 1$. It is clear from Equation (11) that only in this case of $\gamma = 1$ does $t' = t$.

To reiterate, after time t relative to O , O' has moved a distance $x = vt$. The same passage of time, t' , as recorded by O' 's clock relative to O is then given by Equation (11) with $x = vt$, so that

$$t' = \gamma(t - (1 - 1/\gamma^2)t) = t/\gamma \quad (12)$$

But in this case, $t' = \tau$, the *proper time* recorded by O' 's clock, so that Equation (12) must be the *same* as the time-dilation formula, Equation (2a). Then from Equation (12) with $t' = \tau$,

$$\tau = t/\gamma = (1 - v^2/c^2)^{1/2} t \Rightarrow \gamma = (1 - v^2/c^2)^{-1/2} \quad (13)$$

This value of γ is known as the *Lorentz factor* in Special Relativity. With this value of γ , $1 - 1/\gamma^2 = v^2/c^2$, and so Equation (11) reads

$$t' = \gamma(t - vx/c^2) \quad (14)$$

Since, in this scenario involving the observers O and O' , positions in the y and z directions are not affected by relative motion in the x direction only, it follows that $y' = y$ and $z' = z$. These equations, together with Equations (9), (13) and (14) constitute the *special Lorentz transformation* for speed v . To summarise, in the *standard situation* involving observers O and O' described at the beginning of this section, if

(t, x, y, z) is an *event* as recorded by O and (t', x', y', z') is the same event relative to O' , then

$$x' = \gamma(x - vt), t' = \gamma(t - vx/c^2), y' = y, z' = z, \gamma = (1 - v^2/c^2)^{-1/2} \quad (\text{SLT})$$

This set of equations is known as the **special Lorentz transformation** for speed v .

The word ‘special’ in this context refers to the fact that observer O' moves in the chosen x direction relative to observer O . This *standard situation* may be generalised to the case in which O' travels at constant velocity in any direction relative to O . It can also be generalised still further to the case where the Cartesian axes of each inertial frame rotate. However, in practice, given two inertial observers, we may always choose non-rotating Cartesian axes in each observer’s frame so that the standard situation and hence (SLT) holds.

5. Length contraction again

We now revisit the phenomenon of length contraction as an example of applying the special Lorentz transformation. Consider the standard situation involving inertial observers O and O' , as described in the last section. Let a ‘measuring rod’ of length L' relative to O' be positioned along the x' -axis, with one end at the origin in O' ’s inertial frame. Let the length of the rod be recorded as L by O when O' passes O at time $t = 0$. It then follows by (SLT), with $x' = L'$ and $x = L$ when $t = 0$, that

$$x' = L' = \gamma(x - vt) = \gamma(L - v \times 0) = \gamma L$$

$$\therefore L = L'/\gamma = (1 - v^2/c^2)^{1/2} L' \quad (15)$$

Since $v < c$, it follows by Equation (15) that $L < L'$ for $v > 0$. Hence the length of the rod carried by O' is shorter relative to O at $t = 0$. Notice that if, instead, the rod is stationary in O' ’s frame, then its length relative to O' when O passes O' at time $t' = 0$, is given by

$$L' = (1 - (-v)^2/c^2)^{1/2} L = (1 - v^2/c^2)^{1/2} L$$

This result does not contradict Equation (15) since it refers to a different physical situation. Remember that the contraction in length is only a *relative* effect and not an absolute one. It must also be noted that observer *O* may not actually *see* a shortening of the rod in this sense in relative motion. In order to investigate what is seen, the effect of the relative motion on the visual appearance of the moving rod has to be taken into account.

This simple example, demonstrating an application of (SLT), can be generalised to the case where the rod is inclined to the x' -axis. In this case, there is still a length contraction effect but it is no so great as when the rod lies in the direction of motion.

Due to the present state of our technology, no experimental verification of the prediction of length contraction in this sense has been performed, since any such experiment would have to involve accelerating a macroscopic body of an appreciable length to a significant fraction of c .

In the next lecture, I shall examine some more applications of (SLT), within the context of our Normal Realist approach to uniform motion.

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